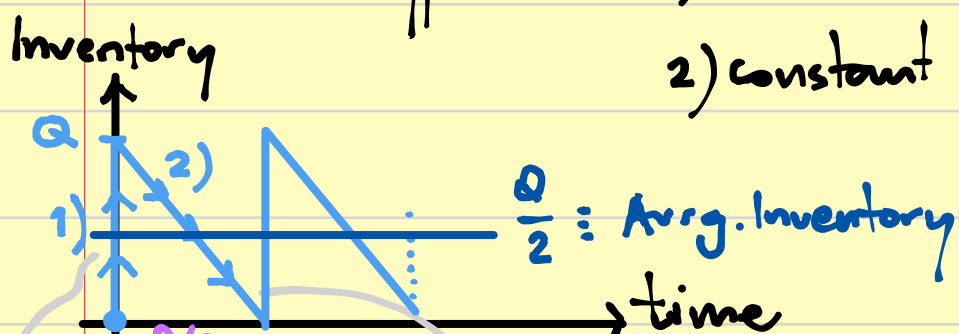




Economic Order Quantity (EOQ)

EOQ I. Hypothesis: 1) immediate delivery & production
2) constant demand



This line is vertical because we have immediate prod & delivery.

This has a constant slope because demand is constant.

Time to deplete the Q inventory is $\frac{Q}{D}$ = Frequency of delivery

Parameters:

D : Demand [Parts/Time]

c : Cost [€/Time]

A : Setup Cost [€]

h : Inventory holding cost [€/Time.Part]

COST FUNCTION:

$$Y(Q) = \text{cost of inventory} + \text{setup cost} + \text{Production cost}$$

$$= h \cdot \frac{Q}{2} + A \cdot \frac{1}{\frac{Q}{D}} + c \cdot D$$

Inventory holding cost per unit
Avg. Inventory
Setup cost
Freq. of Setup
Cost
Demand

The cost of inventory increases linearly with the inventory. The higher the constant the EOQ, the less we setup the machine.

We look for the Economic Order Quantity (Q^*) that minimizes the cost $Y(Q)$.

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$$Y(Q) = \frac{hQ}{2} + \frac{AD}{Q} + cD$$

$$\left. \frac{dY(Q)}{dQ} \right|_{Q=Q_I^*} = 0 \rightarrow \frac{h}{2} + \frac{(-AD)}{Q^2} \Big|_{Q=Q_I^*} = 0 \rightarrow \frac{h}{2} - \frac{AD}{Q_I^{*2}} = 0$$

$$Q_I^* = \sqrt{\frac{2AD}{h}}$$

$$\rightarrow Y(Q_I^*) = \frac{h \cdot Q_I^*}{2} + \frac{AD}{Q_I^*} + cD$$

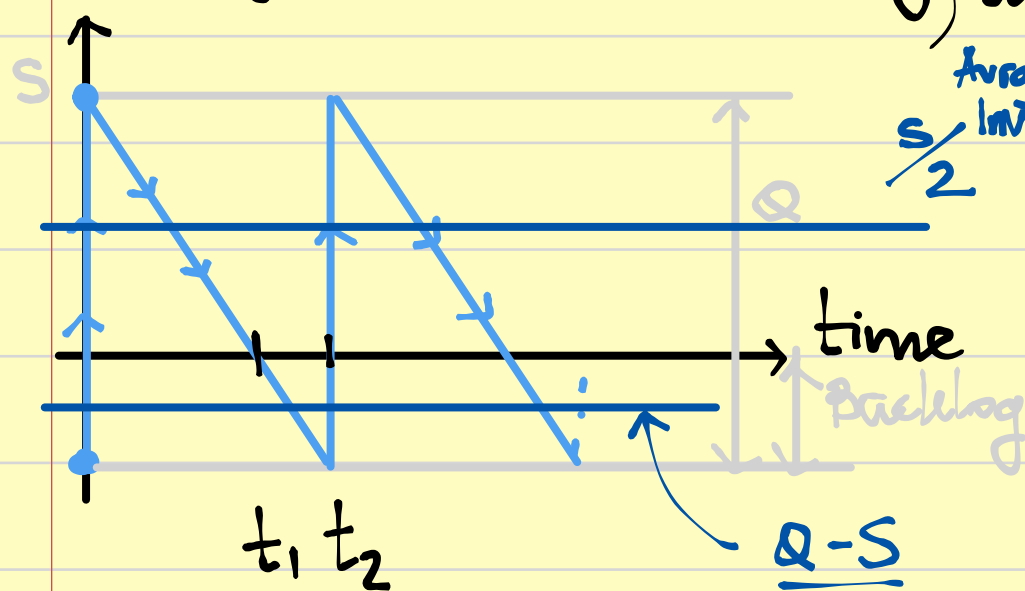
$cD \approx 0$

$$Y(Q_I^*) = \frac{h \cdot \sqrt{\frac{2AD}{h}}}{2} + \frac{AD}{\sqrt{\frac{2AD}{h}}} \rightarrow Y(Q_I^*) = \sqrt{2ADh}$$

EOQ_{II}. Supplier Model.

- Hypothesis:
- 1) we accept "stock out" in which supply has a backlog of orders.
 - 2) delivery & production are immediate
 - 3) demand is constant.

Inventory



PARAMETERS:

p. cost per not supplied orders

[€/unit]

s. Inventory after delivery [unit]

BACKLOG: $Q - s$

$$t_1 = \frac{S}{D} \quad t_2 = \frac{Q}{D}$$

Average Backlog

Cost Function = $Y(Q, S)$ = Inventory holding Cost + Backlog Cost + Setup Cost + Production Cost

Inventory holding cost: $h \cdot \frac{S}{2} \cdot \frac{S}{D} \cdot \frac{D}{Q}$

Annotations:
 - h : holding cost per unit
 - $\frac{S}{2}$: Average Inventory
 - $\frac{S}{D}$: Time to consume inventory
 - $\frac{D}{Q}$: Frequency (how often)

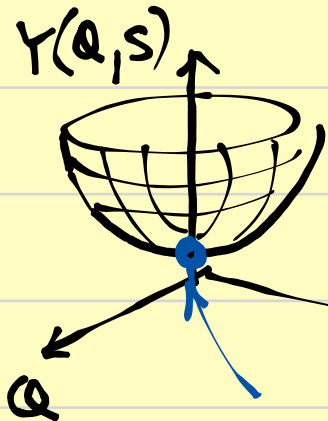
Backlog cost = $p \cdot \frac{Q-S}{2} \cdot \frac{D}{Q}$

Annotations:
 - p : backlog cost
 - $\frac{Q-S}{2}$: Average Backlog
 - $\frac{D}{Q}$: Frequency

Setup Cost: $\frac{AD}{Q}$

Production Cost = cD

$Y(Q, S) = \frac{hS}{2} \cdot \frac{S}{D} \cdot \frac{D}{Q} + p \cdot \frac{Q-S}{2} \cdot \frac{D}{Q} + \frac{AD}{Q} + cD$



$\frac{\partial Y}{\partial Q} \bigg|_{Q=Q^*, S=S^*} = 0$

$\frac{\partial Y}{\partial S} \bigg|_{Q=Q^*, S=S^*} = 0$

$\frac{pD}{2} \left[1 - \frac{S}{Q} \right]$

$\frac{\partial Y}{\partial Q} \bigg|_{Q^*, S^*} = -\frac{hS^{*2}}{Q^{*2}} + \frac{pD}{2} \cdot \frac{S^*}{Q^{*2}} - \frac{AD}{Q^{*2}} = 0$

$\frac{\partial Y}{\partial S} \bigg|_{Q^*, S^*} = \frac{2hS^*}{2Q^*} - \frac{pD}{2Q^*} = 0$

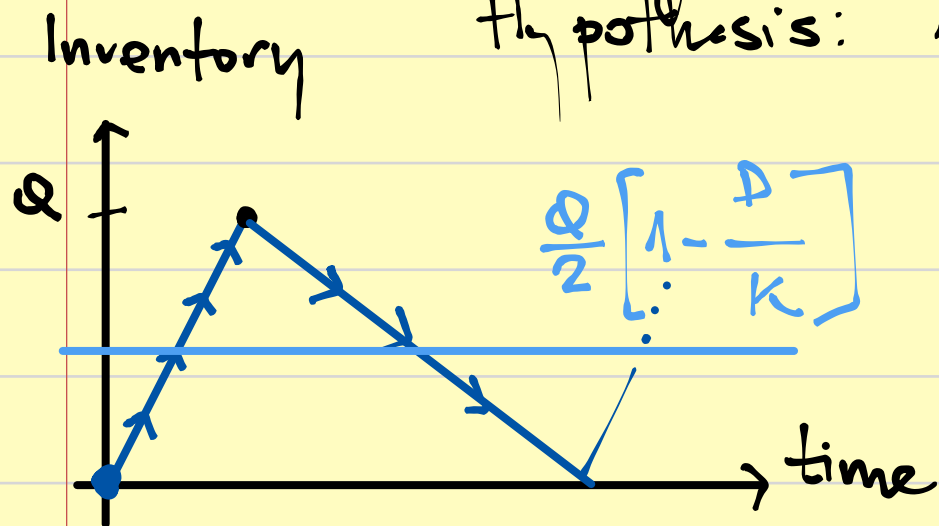
$S_{II}^* = \sqrt{\frac{2AD}{p}} \cdot \sqrt{\frac{p}{p+h}} = Q_I^* \sqrt{\frac{p}{p+h}} \leq Q_I^*$

$$Q_{II}^* = \sqrt{\frac{2AD}{h}} \cdot \sqrt{\frac{p+h}{p}} = Q_I^* \sqrt{\frac{p+h}{p}} > Q_I^*$$

EOQ III. Manufacturing Model. Without Backlog.

the hypothesis:

- 1) constant demand.
- 2) no backlog
- 3) production & delivery are NOT immediate



Parameter:

k : production rate

$Y(Q)$ = holding cost of inventory + setup cost + production cost

$$= h \cdot \frac{Q}{2} \left[1 - \frac{D}{k} \right] + \frac{AD}{Q} + cD$$

$$\frac{dY(Q)}{dQ} \Big|_{Q=Q_{III}^*} = 0 \rightarrow \frac{hQ_{III}^*}{2} \left[1 - \frac{D}{k} \right] - \frac{AD}{Q_{III}^{*2}} = 0$$

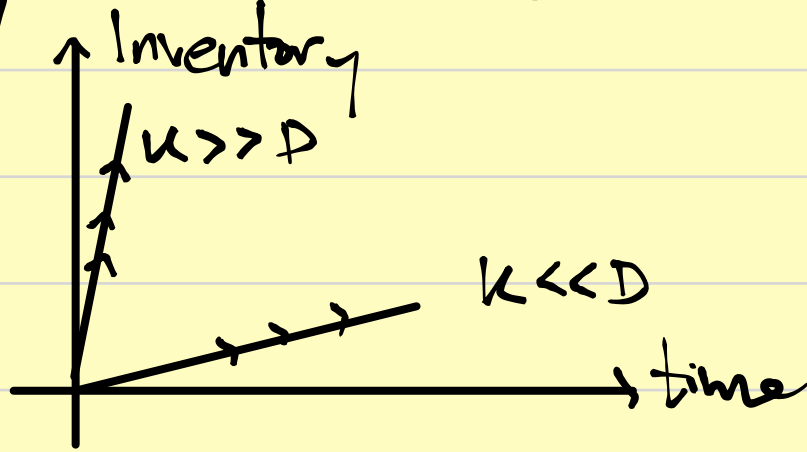
$$Q_{III}^* = \sqrt{\frac{2AD}{h \left(1 - \frac{D}{k} \right)}}$$

Note that if the production rate is infinite $k \gg D$ (this means we are "super fast" producing compared to demand) then $1 - \frac{D}{k} \approx 1 \rightarrow Q_{III}^* \approx Q_I^*$.

$$Y(Q_{III}^*) = \sqrt{2Ah \left(1 - \frac{D}{k} \right)}$$

Again if $k \gg D \rightarrow Y(Q_{III}^*) \approx Y(Q_I^*)$

Conclusion: the more flexible the production, the bigger k , the better we can adapt to demand changes.



• Factory Physics. 

